

A Closed-Form Conservation-Correction Layer for Error-Bounded Lossy Compression of Scientific Fields

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Abstract

Error-bounded lossy compressors such as ZFP and SZ guarantee a pointwise error bound but do not preserve integrated quantities—total energy, mass, momentum—so downstream analyses that check a conservation law observe a bias that grows with the compression ratio. We describe a closed-form correction layer that restores exact conservation of a chosen global quadratic invariant on top of any lossy backend. After the backend reconstructs a field, a single multiplicative factor is computed in closed form so that the reconstruction’s total energy equals the original’s; the factor is stored (4 B per field) and applied on decode. The correction leaves the backend’s compression ratio and pointwise error bound effectively unchanged. On a synthetic three-dimensional turbulence proxy, applied to ZFP across compression ratios from $14\times$ to $60\times$, the layer reduces the relative energy-conservation error by four to six orders of magnitude (from 4.8×10^{-4} to 7×10^{-9} at $60\times$), and the reduction increases with the compression ratio. We give the derivation, the extension to multiple simultaneous invariants, and the method’s limitations. Reference code is released under the Apache-2.0 license.

1 Introduction

The output volume of numerical simulations and instruments has made lossy floating-point compression a standard element of scientific data pipelines. Error-bounded compressors such as ZFP [1] and SZ [2] provide a guaranteed pointwise (per-element) error bound, which controls the local fidelity of a reconstructed field. They do not, however, provide any guarantee on *integrated* quantities computed from that field. Many downstream scientific diagnostics depend on such integrals: total kinetic energy, mass, and momentum are conservation checks, and their drift under compression is a systematic bias rather than bounded noise.

This work addresses that gap with a minimal construction. Given any lossy backend, we compute in closed form the single scale factor that makes the reconstruction’s total energy match the original, store that factor, and apply it on decode. The contribution is deliberately narrow:

- it is not a new codec and does not improve the compression ratio;
- it is a backend-agnostic post-processing layer, demonstrated here on ZFP;
- it converts a chosen global quadratic invariant from “approximately preserved” to “exact to floating-point precision,” at a storage cost of 4 B per field and $O(N)$ arithmetic.

2 Related work

Error-bounded lossy compression for scientific data is an active area; ZFP [1], SZ [2], and TTHRESH [3] are representative and differ in transform and error-control strategy. Prior work has also considered preservation of derived quantities and quantities of interest during compression. The present layer is positioned as the simplest form of that idea: a closed-form, backend-agnostic correction of a single global quadratic invariant applied as post-processing.

3 Method

Let $f \in \mathbb{R}^N$ be a scalar field and let $\hat{f} = C(f; \tau)$ be its reconstruction under a lossy backend C at tolerance τ . Define the energy functional

$$E(x) = \sum_{i=1}^N x_i^2, \quad (1)$$

which is proportional to kinetic energy for a velocity component. We seek a scalar α such that $E(\alpha\hat{f}) = E(f)$. Since $E(\alpha\hat{f}) = \alpha^2 E(\hat{f})$, the solution is closed form:

$$\alpha = \sqrt{\frac{E(f)}{E(\hat{f})}}. \quad (2)$$

The encoder stores α alongside the backend stream; the decoder returns $\alpha\hat{f}$. By construction $E(\alpha\hat{f}) = E(f)$ to floating-point precision, independent of τ and of the backend.

Effect on the pointwise bound. Write $\alpha = 1 + \varepsilon$. From (2), $\varepsilon = \frac{1}{2} (E(f) - E(\hat{f})) / E(\hat{f}) + O(\varepsilon^2)$, so $|\varepsilon|$ is of the order of the backend’s relative energy error, which is small. The corrected pointwise error satisfies

$$\|f - \alpha\hat{f}\|_\infty \leq \|f - \hat{f}\|_\infty + |\varepsilon| \|\hat{f}\|_\infty, \quad (3)$$

i.e. the backend bound is preserved up to a bounded, small additive term.

Vector fields. For a velocity field with components (u, v, w) , a per-component factor α_c conserves each component energy $E(f_c)$ and therefore the total kinetic energy $\frac{1}{2} \sum_c E(f_c)$. The per-field overhead is 4 B per component.

Multiple invariants. Two invariants can be enforced jointly. To preserve both mass $M(x) = \sum_i x_i$ and energy $E(x)$, apply an affine map $x' = a\hat{f} + b\mathbf{1}$ and solve the two constraints $M(x') = M(f)$, $E(x') = E(f)$ for (a, b) ; this reduces to a scalar quadratic. Energy together with vector momentum leads to a small constrained least-squares problem. These extensions are stated here as directions and are not evaluated below.

4 Experimental setup

The test field is a synthetic turbulence proxy on a 64^3 grid: a Taylor–Green base flow superposed with multi-octave sinusoidal noise and a small number of localized Gaussian bursts, stored as `float32`, evaluated over three random seeds. The backend is ZFP (via the `zfp` binding) in fixed-accuracy mode. For each configuration we report the compression ratio against the raw `float32`

array, the maximum absolute pointwise error, and the relative energy drift $|E(\hat{f}) - E(f)|/E(f)$, with and without the correction layer. The field is a proxy and not a solution of the Navier–Stokes equations; results on measured simulation data are left to future work.

5 Results

Table 1 reports the mean over three seeds. Applying the correction adds 4B per field, which does not change the reported ratios at this grid size. The maximum pointwise error is unchanged to the reported precision. The relative energy drift falls from the backend’s 10^{-4} – 10^{-6} range to the 10^{-9} – 10^{-10} range. The reduction is larger at higher compression ratio because the corrected field’s energy is pinned to arithmetic precision regardless of the backend’s drift, whereas the backend’s own drift increases as τ grows.

ZFP tol.	ratio	max abs. error	ZFP energy drift	+ layer energy drift
0.02	14.1×	2.5×10^{-3}	5.5×10^{-6}	3.4×10^{-10}
0.1	22.2×	9.3×10^{-3}	2.5×10^{-5}	2.9×10^{-10}
0.3	35.9×	3.3×10^{-2}	7.1×10^{-5}	1.3×10^{-10}
1.0	59.7×	1.1×10^{-1}	4.8×10^{-4}	7.1×10^{-9}

Table 1: Compression ratio, maximum absolute pointwise error, and relative energy drift for ZFP with and without the correction layer, on a 64^3 synthetic turbulence proxy (mean over three seeds).

6 Applications

Two settings motivate the layer. The first is archival and cloud storage of simulation output. Aggressive lossy compression reduces storage and egress cost, but conservation-sensitive workflows often avoid it because the resulting bias in integrated quantities is uncontrolled. The layer removes that objection for one class of invariant at a fixed, negligible cost, so a conservation check run on the decompressed field returns the original value. The second is in-loop use inside memory-bandwidth-bound solvers. Because the correction touches only decompressed arrays, it can be applied within a solver step rather than only to output files. Lattice-Boltzmann methods are a representative case: their throughput is limited by the traffic of streaming distribution functions, so compressing that traffic while holding the physical invariant fixed is a route to larger single-node domains. This direction is noted here and not benchmarked.

7 Limitations

The evaluation uses a synthetic proxy on a single grid size; measured simulation data and a standard reduction benchmark (e.g. SDRBench [4]) are not yet included. The layer enforces one quadratic invariant (energy) and does not improve non-enforced invariants: a global energy rescale perturbs linear sums, so momentum drift is unchanged or slightly increased. The correction is a single global scalar; it fixes the aggregate, not the spatial distribution of error. Results are shown for one backend (ZFP); backend independence follows from the construction but is demonstrated once. The multi-invariant extension of Section 3 is derived but not evaluated.

8 Conclusion

A closed-form scalar correction makes an error-bounded lossy compressor conserve a chosen global quadratic invariant exactly, at 4 B per field and no change to the compression ratio or the pointwise error bound. On a synthetic turbulence proxy the relative energy drift of ZFP is reduced by four to six orders of magnitude, with the reduction growing as compression becomes more aggressive. The immediate next steps are evaluation on measured data and a standard benchmark, comparison against quantity-of-interest-preserving methods, and evaluation of the multi-invariant extension.

Code and data availability

Reference code and the benchmark used here are available at <https://github.com/alikamp/conserved-field-compression> under the Apache-2.0 license; the reported numbers reproduce from the included notebook.

References

- [1] P. Lindstrom. Fixed-rate compressed floating-point arrays. *IEEE Transactions on Visualization and Computer Graphics*, 20(12):2674–2683, 2014.
- [2] S. Di and F. Cappello. Fast error-bounded lossy HPC data compression with SZ. In *IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 730–739, 2016.
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